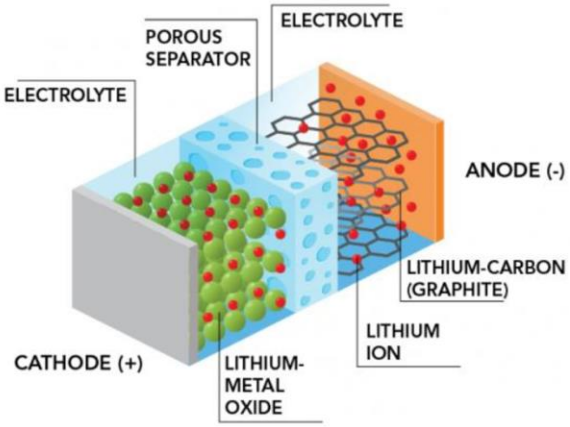
Lithium-ion Battery

Lithium-ion battery: A rechargeable battery that stores and releases energy by moving lithium ions back and forth between a graphite anode and a metal-oxide cathode through a liquid electrolyte, offering high energy density and long cycle life for its size and weight.

Construction

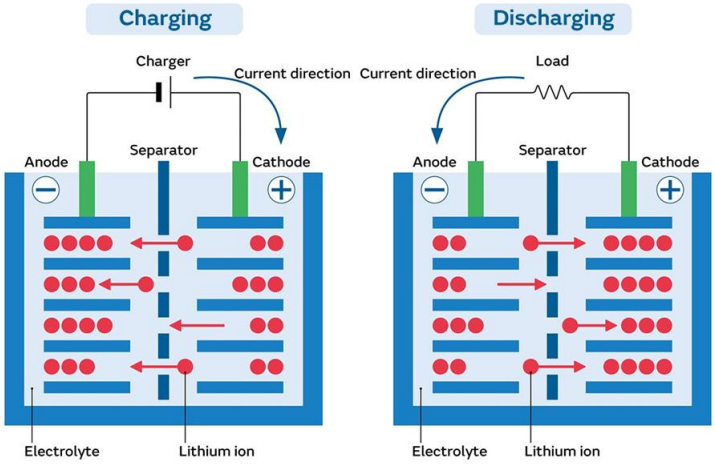


A lithium-ion cell is basically four layers stacked together: (Refer Fig.1)

- Anode (negative electrode) — lithium-graphite (carbon), shown as the honeycomb structure on the right of the image above
- Cathode (positive electrode) — lithium-metal oxide, shown as the green spheres on the left
- Porous separator soaked in electrolyte — the light blue perforated layer sitting between anode and cathode, letting lithium ions (red dots) pass through while keeping the two electrodes from touching
- Current collectors — thin copper foil (anode side) and aluminium foil (cathode side), carrying current out of the cell (not separately labelled in the image, but they sit just outside the electrode layers)

The whole stack sits sealed inside a metal or pouch casing. In the image, you can see the red lithium-ion particles distributed inside both electrode structures — that's lithium sitting stored inside the material, ready to move.

Working Principle



Lithium ions move back and forth between the two electrodes, as shown above:

- **Charging (left panel):** the charger drives ions from the cathode → across the separator → into the anode, where they get stored. You can see the anode filling up with red ion dots as charging happens.
- **Discharging (right panel):** when the battery is powering a load (a motor, a device), ions travel back from anode → across the separator → to the cathode. The movement of lithium ions inside the battery causes electrons to flow through the external circuit, producing electrical energy for the load.
- This back-and-forth motion is why it's often called a "rocking chair" battery — ions simply rock between the two sides depending on whether you're charging or using the battery.

Characteristics

- High energy density — stores a lot of energy for its size and weight
- No memory effect — can be charged anytime without losing capacity
- Low self-discharge — holds charge well when sitting unused
- Performs best in moderate temperatures; Performance decreases at very high or very low temperatures.

Advantages

- High energy density provides longer driving range in EVs.
- Charges relatively fast
- Long lifespan (many charge/discharge cycles)
- Lightweight compared to older battery types (lead-acid, NiMH)

Limitations

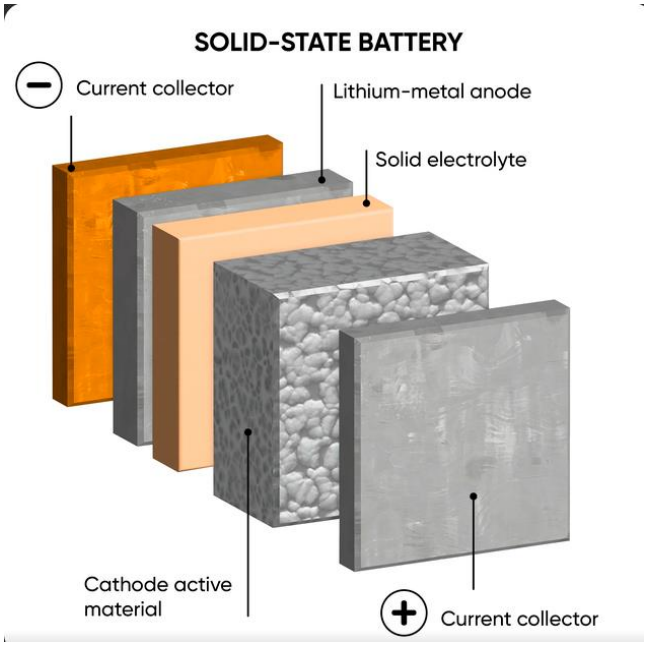
- Liquid electrolyte is flammable — fire risk if damaged or overcharged
- Capacity slowly fades with age and use
- Requires a Battery Management System (BMS) for safe operation and cell protection.
- Reduced range/performance in cold weather

Solid-State Battery

Definition

A **solid-state battery** is a rechargeable battery that uses a **solid electrolyte (such as ceramic, glass, or solid polymer)** instead of a liquid electrolyte to transport lithium ions between the anode and cathode, making it **safer and offering the potential for higher energy density** than a conventional lithium-ion battery.

Construction



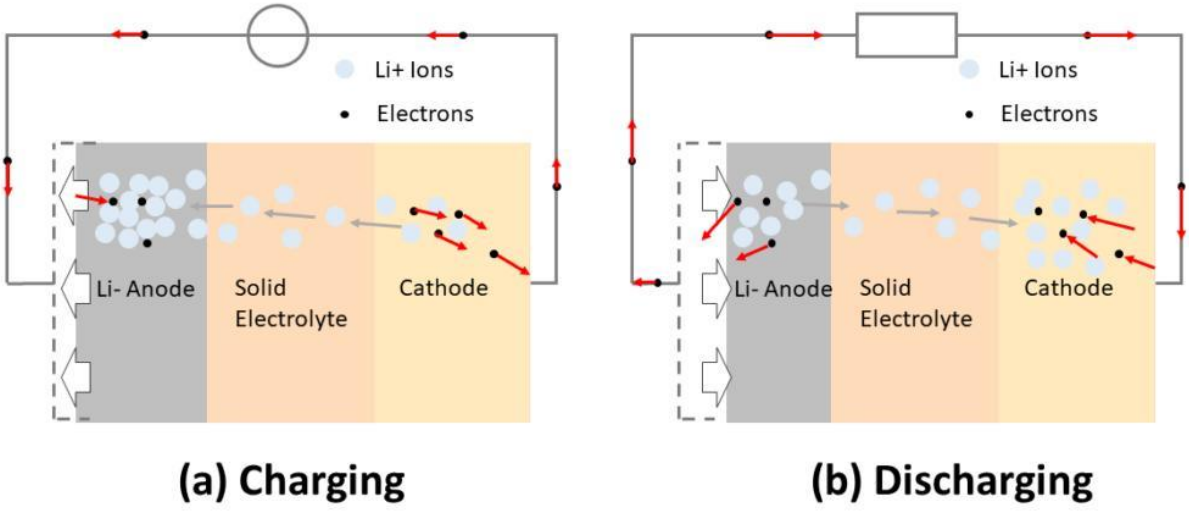
A solid-state battery is built on the same basic principle as a lithium-ion battery, but **the liquid electrolyte and porous separator are replaced by a solid electrolyte**.

From the image above, the layers are:

- **Negative current collector** — **usually copper (Cu) foil;** carries current to and from the anode.
- **Lithium-metal anode** — replaces the graphite anode used in conventional lithium-ion batteries; stores lithium directly as metal, **providing much higher energy density.**
- **Solid electrolyte** — a **ceramic, glass, or solid polymer layer** between the anode and cathode. It **conducts lithium ions and also acts as the separator**, replacing both the liquid electrolyte and porous separator used in lithium-ion batteries.
- **Cathode active material** — the positive electrode where lithium ions are stored and released during battery operation.
- **Positive Current collector** — **usually aluminium (Al) foil;** carries current to and from the cathode.

Unlike conventional lithium-ion batteries, there is no liquid electrolyte inside the battery. The solid electrolyte performs both the functions of ion conduction and electrical insulation between the electrodes.

Working Principle



The figure illustrates the **charging and discharging processes** of a solid-state battery. During operation, **lithium ions (Li^+)** move through the **solid electrolyte**, while **electrons (e^-)** flow through the **external circuit**. The solid electrolyte conducts lithium ions but blocks electrons.

(a) Charging

- An external power source supplies electrical energy to the battery.
- **Lithium ions (Li^+) move from the cathode to the lithium-metal anode through the solid electrolyte.**
- At the same time, **electrons (e^-) flow** through **the external charging circuit to the anode.**
- The lithium-metal anode stores lithium, thereby storing energy in the battery.

(b) Discharging

- When the battery is connected to a load, **lithium ions (Li^+) move from the lithium-metal anode back to the cathode through the solid electrolyte.**
- **Electrons (e^-) flow through the external circuit from the anode to the cathode**, supplying electrical energy to the load.
- This simultaneous movement of lithium ions and electrons produces electric current until the battery is discharged.

Characteristics

- Higher energy density than conventional lithium-ion batteries due to the lithium-metal anode
- Non-flammable solid electrolyte improves battery safety
- Wider operating temperature range
- Potentially longer cycle life (still under evaluation for large-scale commercial applications)
- Lower self-discharge compared to conventional lithium-ion batteries
- Currently expensive and difficult to manufacture consistently on a large scale

Advantages

- Much safer due to the absence of a flammable liquid electrolyte
- Higher energy density, resulting in longer driving range for electric vehicles
- Potential for faster charging due to higher ionic conductivity of advanced solid electrolytes.
- No electrolyte leakage
- Compact cell design since the solid electrolyte also functions as the separator

Limitations

- Very expensive to manufacture at present
- Maintaining good solid-solid interfacial contact between the electrodes and the solid electrolyte is challenging
- Formation of cracks or voids at the electrode–electrolyte interface increases resistance and reduces battery performance.
- Many ceramic solid electrolytes are brittle and difficult to manufacture into thin, defect-free layers
- Still largely in research, development (R&D), and pilot production, with limited commercial availability

Side-by-Side Comparison

Aspect	Lithium-ion	Solid-state
Electrolyte	Liquid (lithium salt in organic solvent)	Solid (ceramic, glass, or solid polymer)
Separator	Separate plastic separator needed	Not needed — solid electrolyte does this job too
Anode material	Usually graphite	Usually lithium metal
Energy density	High	Higher (more energy in the same size/weight)
Safety (fire risk)	Moderate — liquid electrolyte is flammable	Much safer — solid electrolyte doesn't leak or ignite easily
Charging speed	Fast	Potentially faster, with less internal stress
Operating temperature range	Moderate — degrades in extreme heat/cold	Wider — more tolerant of temperature extremes
Cycle life	Long, but capacity fades over years	Expected to be longer (still being proven at scale)
Manufacturing	Mature, established, cost-effective	Difficult and expensive — solid-to-solid contact is hard to make reliably
Availability today	Mainstream — used in almost all current EVs	Mostly R&D / early pilot production
Cost today	Lower, well-established supply chain	High — not yet mass-produced cheaply